

Word Sense Disambiguation (WSD) using CoNLL & WordNet

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Polysemy: Same Word, Different Senses

A word can mean different things depending on the situation in which it is used. For example the word ‘*wooden*’ might be used to talk about the fact that some things are made of wood (‘The chair is wooden’). However, in another situation ‘wooden’ might be used to criticise someone’s acting (‘That actor’s performance was very wooden’). So, what ‘wooden’ means depends on the situation in which it’s used. The same is true for the word ‘*bank*’ in the following two sentences “They had a picnic on the river bank” and “she went to the bank to withdraw cash.”

This feature of words having different meaning is known as **polysemy**. **Word Sense Disambiguation (WSD)** is the natural language processing (NLP) task of determining which meaning of an ambiguous word is intended in a specific context.

Dictionaries

Disambiguation requires two strict inputs: a **dictionary** to specify the senses which are to be disambiguated and a **corpus** to be disambiguated.

One problem with word sense disambiguation is deciding what the senses are, as different dictionaries and thesauruses will provide different divisions of words into senses. Some researchers have suggested choosing a particular dictionary, and using its set of senses to deal with this issue. Generally, however, research results using broad distinctions in senses have been much better than those using narrow ones. Most researchers continue to work on fine-grained WSD.

Most research in the field of WSD is performed by using WordNet as a reference sense inventory for English. WordNet is a computational lexicon that encodes concepts as synonym sets (e.g. the concept of car is encoded as { car, auto, automobile, machine, motorcar }). Other resources used for disambiguation purposes include Roget's

Thesaurus and Wikipedia. More recently, BabelNet, a multilingual encyclopedic dictionary, has been used for multilingual WSD.

Dictionary- and knowledge-based methods

The Lesk algorithm is the seminal dictionary-based method. It is based on the hypothesis that words used together in text are related to each other and that the relation can be observed in the definitions of the words and their senses. Two (or more) words are disambiguated by finding the pair of dictionary senses with the greatest word overlap in their dictionary definitions. For example, when disambiguating the words in "pine cone", the definitions of the appropriate senses both include the words evergreen and tree (at least in one dictionary). A similar approach searches for the shortest path between two words: the second word is iteratively searched among the definitions of every semantic variant of the first word, then among the definitions of every semantic variant of each word in the previous definitions and so on. Finally, the first word is disambiguated by selecting the semantic variant which minimizes the distance from the first to the second word.

An alternative to the use of the definitions is to consider general word-sense relatedness and to compute the semantic similarity of each pair of word senses based on a given lexical knowledge base such as WordNet. Graph-based methods reminiscent of spreading activation research of the early days of AI research have been applied with some success. More complex graph-based approaches have been shown to perform almost as well as supervised methods or even outperform them on specific domains. Recently, it has been reported that simple graph connectivity measures, such as degree, perform state-of-the-art WSD in the presence of a sufficiently rich lexical knowledge base. Also, automatically transferring knowledge in the form of semantic relations from Wikipedia to WordNet has been shown to boost simple knowledge-based methods, enabling them to rival the best supervised systems and even outperform them in a domain-specific setting.

The use of selectional preferences (or selectional restrictions) is also useful; for example, knowing that one typically cooks food, one can disambiguate the word bass in "I am cooking basses" (i.e., it's not a musical instrument).

How does Word Sense Disambiguation (WSD) work in NLP Suite?

Unlike the WordNet aggregation options — which map each word to its *first* (most common) WordNet sense regardless of context — WSD determines the context-correct sense of each word before aggregating. NLP Suite's WSD reads a CoNLL table (produced by any parser: Stanford CoreNLP, Stanza, or spaCy). For each noun and verb it takes the word together with its sentence and applies the Lesk algorithm (Lesk 1986; NLTK implementation), selecting the WordNet synset whose gloss overlaps most with the surrounding words; the word is then aggregated to that synset's semantic category (e.g. the financial-institution sense of *bank* vs. the river-bank sense). This gives more accurate category counts than bare-lemma first-sense

aggregation, at the cost of requiring sentence context — which is why the input is a CoNLL table rather than a word list. WSD is WordNet-based; for VerbNet classes and FrameNet frames use the SRL (Semantic Role Labeling) route, which disambiguates via PropBank/SemLink. Run it from CoNLL Table Analyzer → Advanced CoNLL analyses → Word Sense Disambiguation (WSD).

Word Sense Disambiguation (WSD) vs. Word Sense Induction (WSI): What is the Difference?

Check the TIPS [TIPS_NLP_Word Sense Induction \(WSI\).pdf](#)

Word Sense Induction (WSI). WSD assigns each word to a *known* sense from a fixed inventory (WordNet). Its counterpart, Word Sense Induction (WSI), does the reverse: it discovers how many senses a word takes directly from your corpus — unsupervised, via BERT embeddings + k-means clustering, with no predefined inventory. Use WSD to map words onto established categories; use WSI to find the senses a word actually expresses in your data. WSI is available in the Semantic Analysis GUI (see the *Word Sense Induction (WSI)* TIPS).

What about DBpedia/BabelNet as Disambiguation Options?

Check the TIPS [TIPS_NLP_Annotator DBpedia.pdf](#)

For disambiguating *named entities* (people, places, organizations) rather than common words — or for multilingual WSD — encyclopedic knowledge bases such as DBpedia and BabelNet are used to link mentions to their entries. In NLP Suite, entity linking of this kind is handled by the DBpedia annotator (see the DBpedia TIPS), which complements the WordNet-based WSD described here.

References

- Lesk, Michael. 1986. “Automatic sense disambiguation using machine readable dictionaries: How to tell a pine cone from an ice cream cone (PDF).” *Proc. of SIGDOC-86: 5th International Conference on Systems Documentation. Toronto, Canada*. Archived from the original (PDF) on 2010-08-03.
- Li, Lucy and David Bamman. 2021. “Characterizing English Variation across Social Media Communities with BERT.” *Trans Assoc Comput Linguist* 9:538–556. doi: 10.1162/tacl_a_00383